QuickQueue Smart System

AI-Powered Smart Queue Management

Skip the Line. Wait Anywhere— bringing intelligence, fairness, and real-time visibility to queues in hospitals, banks, and government offices.

TEAM:GLITCH GUILD

The Problem We're Solving

Every day, millions of people waste hours in poorly managed queues. The lack of digital infrastructure in public service environments creates frustration, inefficiency, and inequity.

Long Waiting Lines

Hospitals, banks, and government offices lack systems to manage footfall, leading to chaotic, time-consuming waits.

No Real-Time Tracking

Users have no visibility into queue status — no estimated wait times, no updates, no way to plan ahead.

No Priority Handling

Urgent cases — elderly patients, emergency visitors — get no preferential treatment in standard first-come-first-served systems.

Poor Crowd Management

Without analytics or live monitoring, administrators cannot proactively manage congestion or allocate staff effectively.

Our Solution: QuickQueue

QuickQueue is a smart, end-to-end digital queue management platform that eliminates the chaos of physical waiting lines. Users receive instant tokens online, get real-time wait predictions, and benefit from intelligent priority handling — while admins gain full live control of the entire system.

For Users

- Instant digital token generation
- Real-time waiting time prediction
- Priority-based queue handling
- Voice & notification alerts

For Administrators

- Live queue monitoring dashboard
- Call / Serve system for counters
- Priority highlighting & overrides
- Analytics & queue status control

Key Features

Smart Token Generation

Users receive a unique digital token instantly upon entering the system — no physical tickets, no manual registration.

Real-Time Wait Prediction

ML-powered algorithms calculate accurate wait times based on current queue depth, service rate, and historical data.

Location-Based Services

Integrated OpenStreetMap APIs suggest the nearest available service center with the shortest expected wait.

Priority Queue System

Intelligent logic automatically elevates urgent cases — elderly, disabled, or emergency visitors — ahead of the regular queue.

Voice Alert System

Automated voice announcements notify users when it's their turn, reducing missed calls and improving flow.

Live Admin Dashboard

A centralized control panel gives administrators full visibility and control over queue status, user flow, and analytics.

Technology Stack

QuickQueue is built on a lean, modern stack designed for rapid deployment, real-time performance, and AI integration. Every layer was chosen for simplicity, reliability, and hackathon-ready speed.

Frontend

HTML, CSS, JavaScript — A responsive, lightweight interface that works seamlessly across devices without requiring a framework.

Backend

Flask (Python) — A minimal, flexible web framework that handles API logic, queue management, and real-time data processing.

Database

SQLite — A zero-configuration embedded database ideal for storing tokens, user sessions, and queue state during the demo.

AI Module

ML-Based Prediction — A custom-trained model that forecasts wait times using queue depth, historical patterns, and service rates.

APIs

OpenStreetMap — Provides location data and hospital/service center information for smart, proximity-based routing suggestions.

How It Works

QuickQueue's workflow is designed to be frictionless for users and powerful for administrators. From token generation to live queue management, every step is automated and intelligent.

User Enters Details

Visitor inputs basic information through a simple web interface.

Token Generated

System instantly issues a unique digital token with a queue position.

Queue Auto-Sorted

AI logic prioritizes urgent cases and optimizes queue order dynamically.

Wait Time Calculated

ML model predicts real-time wait duration based on current conditions.

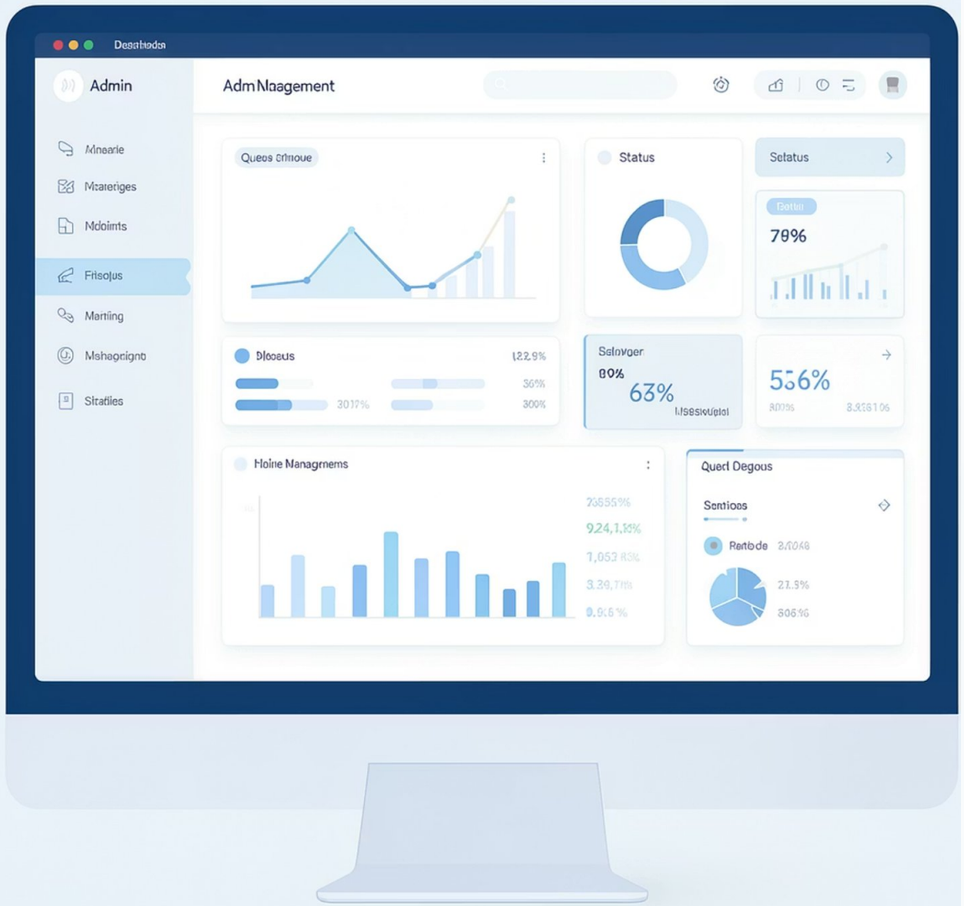
Alerts Triggered

Voice and digital notifications inform users when it's their turn.

Admin Manages Queue

Live dashboard allows staff to monitor, call, and control the queue in real time.

This end-to-end automated pipeline ensures zero manual intervention from token issuance to service completion — reducing human error and dramatically improving throughput.



Admin Dashboard

The Admin Dashboard is the command center of QuickQueue — giving staff complete, real-time control over queue operations. Designed for clarity and speed, it enables a single administrator to manage an entire service floor efficiently.

👁 Live User Monitoring

See every active token, current position, and wait status in real time on a single unified view.

📢 Call / Serve System

One-click call functionality alerts the next user via voice announcement and on-screen display.

⭐ Priority Highlighting

Urgent cases are visually highlighted, ensuring staff never miss a priority visitor who needs immediate attention.

📊 Analytics & Charts

Historical and live data visualizations help administrators identify peak hours, bottlenecks, and staffing needs.

What Makes QuickQueue Innovative

Beyond basic queue management, QuickQueue introduces several intelligent features that set it apart from traditional token systems — combining AI, real-time data, and smart automation.

🤖 AI-Based Wait Prediction

Unlike static estimates, our ML model continuously learns from live queue data to deliver accurate, dynamic wait time forecasts that improve over time.

🧠 Smart Priority Logic

The system automatically detects and elevates priority cases based on user-defined criteria — ensuring fairness without manual intervention.

🔊 Voice Alert Notifications

Integrated voice synthesis announces user tokens in real time, reducing missed calls and improving accessibility for visually impaired users.

📍 Location-Based Hospital Suggestions

Using OpenStreetMap APIs, QuickQueue recommends the nearest facility with the shortest queue — distributing load intelligently across locations.

Measurable Impact

QuickQueue is designed to deliver tangible improvements across every dimension of public service operations — from individual wait times to system-wide efficiency.

40–60%

Wait Time Reduction

AI-optimized queue routing and real-time predictions significantly cut average waiting times for all users.

100%

Digital Transformation

Replaces paper-based, manual queue systems with a fully digital, automated, and trackable experience.

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Manual Token Handling

Complete automation from token generation to notification eliminates human error and staff overhead.

QuickQueue improves public service efficiency, reduces crowd congestion, and creates a more dignified experience for every visitor — whether in a hospital, bank, or government office.

The Road Ahead

QuickQueue is just getting started. Our roadmap envisions a fully scalable, mobile-first platform ready for city-wide and government-level deployment.

1

Mobile App

Native iOS & Android apps for on-the-go token management and push notifications.

2

QR Code Entry

Scan-to-join system at physical locations for seamless walk-in integration.

3

AI Crowd Prediction

Forecast peak hours and proactively manage staffing and resource allocation.

4

Government Deployment

Pilot programs with municipal offices and public hospitals for real-world validation.

5

Multi-City Expansion

Scale the platform across cities and states, creating a unified smart queue network.

QuickQueue Smart System — Thank you for your time. We're ready for your questions!